Chapter 2 • Logarithms

Solved Exercise 2.4

1. Evaluate without a calculator.

(i) $\log_2 18 - \log_2 9 = \log_2(18/9)$
$= \log_2 2$
$= 1.$
(ii) $\log_2 64 + \log_2 2 = 6 + 1$
$= 7.$
(iii) $(1/3)\log_3 8 - \log_3 18 = \log_3 2 - \log_3 18$
$= \log_3(\frac{2}{18})$
$= \log_3(\frac{1}{9})$
$= -2.$
(iv) $2\log 2 + \log 25 = \log(2^2 \times 25)$
$= \log 100$
$= 2.$
(v) $(1/3)\log_4 64 + 2\log_5 25 = (1/3)(3) + 2(2)$
$= 1 + 4$
$= 5.$
(vi) $\log_3 12 + \log_3 0.25 = \log_3(12 \times 0.25)$
$= \log_3 3$
$= 1.$

2. Write each expression as a single logarithm.

(i) $(1/2)\log 25 + 2\log 3 = \log 5 + \log 9$
$= \log 45.$
(ii) $\log 9 - \log(1/3) = \log[\frac{9}{\frac{1}{3}}]$
$= \log 27.$
(iii) $\log_5(b^2) \cdot \log_a(5^3) = 2\log_5 b \cdot 3\log_a 5$
$= 6\log_a b$, by change of base.
$6\log_a b$
(iv) $2\log_3 x + \log_3 y = \log_3(x^2 y).$
(v) $4\log_5 x - \log_5 y + \log_5 z = \log_5(x^4 z / y).$
(vi) $2\ln a + 3\ln b - 4\ln c = \ln(a^2 b^3 / c^4).$

3. Expand using the laws of logarithms.

(i) $\log(11/5) = \log 11 - \log 5.$
(ii) $\log_5 \sqrt{(8a^6)} = (1/2)(\log_5 8 + 6\log_5 a)$
$= (\frac{3}{2})\log_5 2 + 3\log_5 a.$
(iii) $\ln(a^2 b/c) = 2\ln a + \ln b - \ln c.$
(iv) $\log(xy/z)^{\frac{1}{9}} = (\frac{1}{9})(\log x + \log y - \log z).$
(v) $\ln \sqrt[3]{(16x^3)} = (1/3)(\ln 16 + 3\ln x)$
$= (\frac{4}{3})\ln 2 + \ln x.$
(vi) $\log_2[\frac{1-a}{b}]^5 = 5[\log_2(1-a) - \log_2 b].$
$5[\log_2(1-a) - \log_2 b]$. Domain: every original logarithm argument must be positive.

4. Solve the six equations for x.

(i) $\log 2 + \log x = 1$
$\Rightarrow \log(2x) = 1$
$\Rightarrow 2x = 10$
$\Rightarrow x = 5.$
(ii) $\log_2 x + \log_2 8 = 5$
$\Rightarrow \log_2(8x) = 5$
$\Rightarrow 8x = 32$
$\Rightarrow x = 4.$
(iii) $81^x = 243^{x+2}$
$\Rightarrow 3^{4x} = 3^{5(x+2)}$
$\Rightarrow 4x = 5x + 10$

studysapiens.com

$\Rightarrow x=-10.$
(iv) $(1/27)^{x-6}=27$
$\Rightarrow 3^{-3(x-6)}=3^3$
$\Rightarrow -3x+18=3$
$\Rightarrow x=5.$
(v) $\log(5x-10)=2$
$\Rightarrow 5x-10=100$
$\Rightarrow x=22$
domain requires $x>2$.
22
(vi) $\log_2(x+1)-\log_2(x-4)=2$
$\Rightarrow \frac{x+1}{x-4}=4$
$\Rightarrow x+1=4x-16$
$\Rightarrow x=\frac{17}{3}$
domain requires $x>4$.
$\frac{17}{3}$ All solutions satisfy their original domains.

5. Evaluate with logarithm tables.
(i) $N=\frac{3.68\times4.21}{5.234}$: $\log N\approx0.5658+0.6243-0.7188=0.4713$
$\Rightarrow N\approx2.960.$
(ii) $N=4.67\times2.11\times2.397$: $\log N\approx0.6693+0.3243+0.3797=1.3733$
$\Rightarrow N\approx23.62.$
(iii) $N=\frac{20.46^2\times2.4122}{754.3}$: $\log N\approx2(1.3109)+0.3824-2.8775=0.1267$
$\Rightarrow N\approx1.339.$
(iv) $N=\frac{\sqrt[3]{9.364\times21.64}}{3.21}$: $\log N\approx\frac{0.9715}{3}+1.3353-0.5065=1.1526$
$\Rightarrow N\approx14.21.$
14.21. Small last-digit differences may arise from table rounding.

6. For $M = \log_{10}(\frac{A}{A_0})$, A = 10,000 and A_0 = 10, find the earthquake magnitude.

$M = \log_{10}(\frac{10000}{10})$
$= \log_{10}1000$
$= 3.$
M = 3.

7. An investment $y = 100{,}000(1.05)^t$ doubles. Find t.
$2 = (1.05)^t$
$\Rightarrow t = \log \frac{2}{\log 1.05} = 14.2067...$
Approximately 14.21 years; about 14 years to the nearest year.

8. Temperature $T = 20(0.97)^{\frac{h}{100}}$. Find T at h = 500 m.
$T = 20(0.97)^5$
$= 17.174...$
17.17°C approximately.

studysapiens.com